

2008 SEA-DOO · RXT-X 255

100-Hour Supercharger Service Manual

The complete job — parts to order, where to buy, prices, tools, torque specs, and the full step-by-step rebuild.



A

Prepared for Arman — from Shawn (Marcus's father)

A full reference so the whole job is clear before a single bolt is touched.

- 1** At a glance — job, time & total cost
- 2** The machine & the service
- 3** Parts to order (with links & prices)
- 4** Special tools to order
- 5** Full cost breakdown
- 6** Tools checklist
- 7** Step-by-step rebuild
- 8** Torque & spec quick reference
- 9** The plan & two questions

The whole job in one screen

THE SERVICE

Supercharger rebuild

DUE AT

100 hrs / 2 yrs

BENCH TIME

~2–3 hrs

TURNAROUND

~½ day

OEM rebuild kit (P/N 420881102)

\$350–\$500

Special tools

if not owned

\$75–\$130

Oil + filter, Loctite, o-rings

\$50–\$80

Typical parts total

≈ \$475–\$710

Shawn buys all parts and ships them to you. You supply the wrench time.
Labor settled on your terms.

2008 Sea-Doo RXT-X 255

ENGINE

Rotax 1503 4-TEC

POWER

255 HP · S/C

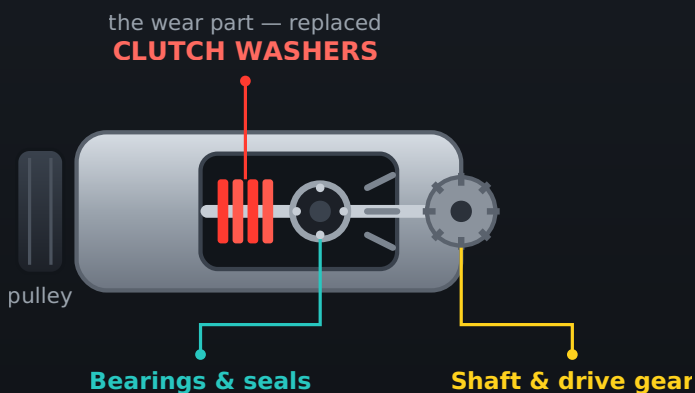
SUPERCHARGER

16-tooth · intercooled

HOURS

~100 hrs

Why the rebuild matters



The clutch washers wear, the clutch slips — and if ignored, debris can reach the engine.

Not optional: worn clutch washers let the clutch slip and can shed metal *into the engine* — a ~\$450 job that prevents a new-motor bill. Plastic bearing cages also go brittle with heat over time (the "2-year" half of the rule).

What we buy — and where

The one required item is the OEM rebuild kit. Below are real vendors, prices, and links. Tap a link to go straight to the product.

① Required — OEM rebuild kit

P/N 420881102 · 16-tooth · steel washers

RECOMMENDED

Sea-Doo / BRP (genuine)

≈ \$390

Factory OEM kit direct from BRP — the exact part, steel washers, IsoFlex spec.

Open BRP product page →

sea-doo-shop.brp.com → Supercharger Repair Kit 420881102

RIVA Racing

\$499.99

Genuine OEM kit, free shipping over \$150. Updates to '08 metal-washer spec.

Open RIVA product page →

rivaracing.com → OEM Sea-Doo 215/255/260 kit 420881102

Offshore Marine Parts

\$350–\$420

OEM kit, fits GTR/GTX/RXP/RXT intercooled 16-tooth.

Open Offshore Marine page →

offshoremarineparts.com → Rebuild Kit, Supercharger (16-tooth)

Amazon (OEM)

\$380–\$470

Genuine BRP kit 420881102 — fast Prime shipping option.

[Open Amazon listing →](#)

[amazon.com](#) → Sea-Doo OEM Supercharger Rebuild Kit 420881102

② Budget alternative — aftermarket kit

PWCParts combo (kit + tools + grease)

≈ \$230

Metal washers, bearings, seals **plus** the holding tools and required IsoFlex grease — one box, nothing else to buy. Replaces 420881102.

[Open PWCParts combo →](#)

[pwcparts.com](#) → PWC-SD-0090-KT (kit + tool kit + grease)

My lean: genuine OEM kit for reliability. But the PWCParts combo is the best value if you'd rather have everything — including the special tools and grease — arrive in a single box. Your call.

The two that crack the nut

The supercharger shaft must be held while the cap nut is removed/torqued. If you don't already have these, I'll add them to the order. (Skip this section entirely if you choose the PWCParts combo above — tools are included.)

RIVA shaft-holder tool

\$74.95

Holds the compressor shaft so you can break the cap nut. Item RS1750-ISH-949.

[Open RIVA shaft-holder →](#)

[rivaracing.com](#) → RIVA Sea-Doo Supercharger Shaft Holder Tool

Gear-holder tool (1503 16-tooth)

≈ \$55

Holds the drive gear during disassembly/reassembly. Precision P/N 950-110-17.

[Open gear-holder tool →](#)

[pwcparts.com](#) → Precision Supercharger Gear Holding Tool 950-110-17

The numbers

OEM rebuild kit (420881102)		\$350–\$500
— or aftermarket combo kit		≈ \$230
RIVA shaft-holder tool	if needed	\$75
Gear-holder tool	if needed	\$55
IsoFlex Topas NB 52 grease	often incl.	\$0–\$25
Engine oil + filter (XPS 4-stroke)		\$45–\$60
Blue Loctite 243 + o-rings		\$15
External intercooler	only if leaking	\$150–\$300
Typical parts (OEM + tools)		≈ \$475–\$710

Labor is separate and on your terms, Arman — this manual is just so the parts and scope are crystal clear.

On the bench

TORQUE

Torque wrench (ft-lb)

HOLD

Bench vise

BEARINGS

Shop / arbor press

CLIPS

Snap-ring pliers

SPECIAL

Shaft + gear holders

LOCK

Blue Loctite 243

GREASE

IsoFlex Topas NB 52

SOCKETS

Metric set + drivers

Grease matters: the clutch washers must be assembled with **IsoFlex Topas NB 52** — not regular grease. Good kits include it; if not, it's on the order.

Start to finish

You know this work, Arman — this is just the full sequence on record, with the key torque call-outs.

A Remove the supercharger

1 Disconnect the battery



Negative first. Pull the seats / rear access panel to reach the engine bay.

2 Clear access



Remove the air intake / silencer and any covers over the PTO (clutch) side where the supercharger sits.

3 Free the charge pipe



Loosen the clamp and pull the intercooler / charge hose off the supercharger outlet.

4 Unbolt & pull the S/C



Remove the retaining bolts and slide the supercharger off its splined drive. Cap the opening so nothing drops in.

B Disassemble on the bench

5 Secure it



Mount the supercharger in the vise (use a support plate / soft jaws). Fit the shaft-holder tool.

6 Remove the cap nut



Hold the shaft, crack the cap nut. (Reassembly spec: **22 ft-lb.**)

🕒 Cap nut - 22 ft-lb

7 Pull the clutch washer stack



Remove the washers in order — note orientation. This is the main wear item being renewed.

8 Off with the drive gear & bearings



Fit the gear-holder, remove the drive gear, then press out the old bearings and seals.

C Rebuild

9 Clean everything



Solvent-clean the housing and impeller. Inspect the volute and shaft bore for scoring.

10 Press in new bearings & seals



From the kit: new bearings (×2), seals, o-ring. Press squarely, fully seated.

11 New washers — with IsoFlex grease



Assemble the new clutch washers in the correct order, greased with **IsoFlex Topas NB 52**. Fit the new shaft & drive gear.

➔ Grease: IsoFlex Topas NB 52 only

D Torque & verify

12 Torque the cap nut

Hold the shaft and torque the cap nut to spec.



⌚ Cap nut · 22 ft-lb

13 Check clutch slip torque

Confirm the rebuilt clutch slips within spec — not too loose, not seized. This is the quality check that protects the engine.



E Reinstall & test

14 Refit the supercharger

New o-ring, slide onto the splined drive, torque the mounting bolts (per shop-manual spec), reconnect the charge pipe & intake.



15 Oil change & battery

Fresh XPS 4-stroke oil + filter is the right move while you're in here. Reconnect the battery.



16 Run it & confirm boost

Start, check for leaks, confirm it pulls boost and holds. A quick water test and it's done.



Done = supercharger rebuilt with new washers/bearings/seals, cap nut at 22 ft-lb, slip torque verified, back on the ski, fresh oil, holding boost. Clean and quiet.

Pin it to the bench

Supercharger cap nut

22 ft-lb

Clutch-washer grease

IsoFlex NB 52

Threadlocker (where spec'd)

Blue 243

S/C mounting bolts

per manual

Slip torque

check to spec

Service interval

100 hr / 2 yr

Cap-nut torque, grease type and interval are well-documented for the 1503 4-TEC. For mounting-bolt torque and the exact slip-torque window, follow the Sea-Doo shop manual — values can vary by year/revision. You're the mechanic; this sheet is a convenience, not a substitute for the manual.

How we run it

1

I order everything

Kit, tools (if needed), grease, oil — shipped to your shop.

2

I drop the ski off

On the trailer, on your schedule. Parts already there.

3

You rebuild

~2–3 hrs bench + pull/reinstall. ~½ day total.

4

I pick up & we settle

Labor on your terms. Done.

Two questions for you

1. Genuine OEM kit, or the all-in-one aftermarket combo (kit + tools + grease)?

2. Do you already have the shaft + gear holder tools, or should I add them?

Answer those two and I'll place the order today and we'll set a drop-off.

Thanks a ton, Arman —
Shawn

Prices are current market ranges and move over time; the links go to live product pages. Reference manual only — always defer to the Sea-Doo shop manual for torque specs and procedures.